Adam Petrovic

Senior Software Engineer – Distributed Systems, Cloud Infrastructure & Reliability

Sydney, Australia

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Summary

Senior software engineer with 14+ years building cloud platforms, object storage, and reliability systems, including 10+ years at Atlassian. Current work is on Atlassian's internal S3-compatible storage service: GCP migration, Kubernetes, storage analytics, incident response, and cost/reliability work. Earlier roles covered observability systems, a $2M+ metrics migration, 180M metrics/minute pipelines, synthetic monitoring, and Jira Cloud SRE. Enjoys turning complexity and risk into RFCs, production code, and tooling adopted by other teams across the organisation.

Skills

- **Languages:** Kotlin, Go, Python, Java, TypeScript, JavaScript, SQL, Bash
- **Cloud & Infrastructure:** AWS, GCP, Kubernetes, Docker, Spinnaker, Flux, Helm, Kustomize, Talos Linux, CI/CD
- **Storage & Data Systems:** S3, GCS, DynamoDB, PostgreSQL, MySQL, Redis, sharding, backup/restore
- **Reliability & Observability:** SignalFx, Datadog, Prometheus, Grafana, Splunk, StatsD, Sentry, synthetic monitoring, SLOs
- **Data & Performance:** Kafka, Apache Flink, SQS, Databricks, Athena, Locust, high-volume telemetry pipelines

Experience

Atlassian (10+ Years) — **Sydney, Australia**

Senior Engineer, Transactional Data Platform — *August 2023 – Present*

- Led GCP migration for Atlassian's internal S3-compatible storage service across compute, credentials, data-integrity checks, and deployment readiness, delivering a critical milestone four weeks early.
- Designed Kotlin architecture for S3-compatible storage APIs, data residency, encryption, backup/restore, GCS integration, and multi-shard deployment paths for cloud-agnostic operation across AWS and GCP.
- Wrote RFCs for GCS checksum handling, reducing control-plane dependencies, and seekable compression, turning reliability, cost, and migration trade-offs into clear storage direction.
- Built S3 Inventory and DynamoDB export ingestion into the data platform, giving service owners storage cost attribution, object verification, and self-service analytics over multi-terabyte datasets.
- Replaced slow storage investigations with self-service queries, cutting 24-hour verification jobs to 5-minute checks and safely enabling $25K/month object-tag savings across production buckets.
- During a high impact incident, built analysis pipelines under pressure to join database exports with S3 inventory, prove no data loss, and start continuous object verification for the platform's top risk.
- Created deployment-descriptor diff pipeline in Go/Bitbucket that flags risky infrastructure changes before merge, adopted by multiple platform teams to catch blast-radius issues before production.
- Led sharding work that split the storage service into isolated deployment units, delivering an isolated-cloud shard three weeks early and a repeatable rollout method for future shards and regions.
- Built repeatable Locust load tests for staging, letting engineers validate feature changes under production-like traffic, compare results across runs, and build confidence before deployment.

Engineering Manager, Observability — *October 2020 – August 2023*

- Led Better Metrics migration from Datadog to SignalFx for Atlassian engineering, saving more than $2M annually while keeping critical metrics workflows stable for thousands of engineers during transition.
- Managed observability engineers through vendor evaluation, roadmap planning, stakeholder communication, incident reviews, hiring, coaching, performance development, and operational trade-offs.

- Led delivery of an HTTP routing layer between metrics aggregators and upstream observability sinks, using Atlassian's open-source GoStatsD project to make the Datadog-to-SignalFx migration safer at scale.

Senior Engineer, Observability *June 2017 – October 2020*

- Built and operated Atlassian's metrics infrastructure at 180M datapoints/minute, supporting production monitoring, alerting, and incident diagnosis across cloud services and internal platforms.
- Developed Pollinator, a Go-based synthetic monitoring and post-deployment verification platform used across regions to catch reliability and performance regressions in release flows before customers did.
- Extended GoStatsD with Prometheus-style histograms and moved telemetry toward Kafka/Flink, improving percentile metrics, stream-processing headroom, and SLO diagnosis for service teams.

Senior Site Reliability Engineer, Jira Cloud *March 2015 – June 2017*

- Led a cross-continental SRE group delivering Jira Cloud's European infrastructure ahead of a public launch milestone, supporting enterprise customers, regional growth, and product commitments.
- Reworked Jira Cloud site imports and introduced Wheel of Misfortune training, reducing failed customer imports, on-call toil, and incident-response gaps through rehearsed failure scenarios.

Freelancer.com (3.5 Years) **Sydney, Australia**
Software Engineer *August 2011 – February 2015*

- Built platform services for authentication, messaging, notifications, email, analytics, and fraud systems in a global marketplace using Python, Go, Redis, RabbitMQ, Thrift, and async workers.
- Helped move critical paths away from PHP monolith patterns into service-oriented APIs and async processing, improving request latency, failure isolation, and delivery speed for core marketplace flows.

Education

The University of Sydney **Sydney, Australia**
Bachelor of Information Technology, Computer Science *2008 – 2012*

- Dean's List with a distinction average.